Hostel Management System

System Validation Report

Validation of the delivered system against SRS v1.0

Project	Hostel Management System (HMS)
Team	Team 3
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Repository	github.com/adarshv15/hostel-management-system-team3 (commit 857e9c1)
Baseline document	Software Requirements Specification v1.0 (TEAM_3_HMS_SRS)
Report version / date	1.0 / 09-11-2026
Status	Draft for team review
Test period	Testing phase (plan: 2 Nov to 6 Nov); this run executed 09-11-2026

1. Purpose and Scope

This report validates the Hostel Management System as built at commit `857e9c1` against the requirements in SRS v1.0: 24 functional requirements (HMS-F), 5 non-functional requirements (HMS-NF) and 5 security requirements (HMS-SR). Its question is whether the delivered system does what the SRS says it should, as distinct from whether individual units behave as coded (see the Unit Test Report).

The SRS note states 18 functional requirements, but sections 4.1 to 4.6 list 24. All 24 listed requirements are assessed here.

2. Validation Method

Technique	What was done
Automated tests	The 32 unit tests were run and mapped to requirements (all passed).
Runtime probes	A separate script drove the running application through Django's test client: anonymous, non-staff, staff and view-only access; password storage; allocation rules; checkout behaviour; invoice creation; audit behaviour (section 5.1).
Configuration review	`settings.py` was inspected and `manage.py check --deploy` was run (section 5.2).
Source inspection	Models, admin classes, views and URLs were read to confirm which SRS features exist.

Status definitions. Met: implemented and evidenced. **Partially met**: some acceptance criteria satisfied, others missing. **Not implemented**: no corresponding feature found. **Not verified**: cannot be judged without a test that was not run (load, survey, production monitoring).

2.1 Environment

Item	Detail
Operating system	Linux 6.18 (cloud sandbox)
Python	3.13.18 (the project CI targets Python 3.12)
Django	5.2.19 (requirements.txt allows >=5.1,<6.0)

Item	Detail
Coverage tool	coverage.py 7.17.2 with `--branch`
Database	SQLite, Django-managed in-memory test database
Test runner	Django `manage.py test` (unittest based, `django.test.TestCase`)
Execution date	09-11-2026

3. Summary

Category	Total	Met	Partially met	Not implemented	Not verified
Functional (HMS-F)	24	4	13	7	0
Non-functional (HMS-NF)	5	0	0	0	5
Security (HMS-SR)	5	2	2	1	0
Total	**34**	**6**	**15**	**8**	**5**

Of the 24 functional requirements, 4 are met, 13 are partially met and 7 are not implemented. Of the 12 High-priority functional requirements, 2 are met, 8 are partially met and 2 (fee structures and the payment gateway) are not implemented.

Overall verdict: the system is a working Sprint 2 foundation for staff-run record keeping (students, rooms, allocations, invoices, payments, complaints, leave, visitors, meals, notices and audit trail through the admin console). It does not yet satisfy the SRS acceptance exit criteria (section 6).

4. Requirement-by-Requirement Validation

4.1 Functional requirements

ID	Requirement	Pri.	Status	Evidence and findings
HMS-F-001	Online student registration with personal, academic and guardian details	High	**Partially met**	Student model holds all fields and defaults to Pending; records are created by staff in the admin console. No public self-registration form (UT-05).
HMS-F-002	Upload ID and admission documents (PDF/JPEG)	Medium	**Not implemented**	No file field or upload handling exists in the models or admin.
HMS-F-003	Warden approves or rejects registration with a reason	High	**Partially met**	Student status supports Active and Rejected. No reason field and no notification to the student.
HMS-F-010	Room and bed inventory with real-time occupancy	High	**Met**	Room model with capacity; admin list shows occupied and available beds (UT-06, UT-07, UT-08).
HMS-F-011	Allocate an available bed to an approved student	High	**Met**	Allocation reduces available beds (UT-08). Probe: a Pending student was rejected with "Only active students can be allocated a bed".
HMS-F-012	Prevent allocation of an occupied bed; show only available beds	High	**Partially met**	Double allocation and over-capacity are rejected by DB constraint and validation (UT-09, UT-25 to UT-27). Bed label is free text, so only available beds are not offered to the user.
HMS-F-013	Student submits a room change request	Medium	**Not implemented**	No transfer-request model or workflow.
HMS-F-014	Warden approves or rejects a transfer and re-allocates	Medium	**Not implemented**	Depends on F-013; not present.
HMS-F-015	Automatic de-allocation on checkout or inactive status	High	**Partially met**	Setting `checked_out_at` frees the bed (probe: free beds = 1). Marking a student Inactive did not release the bed (probe: `checked_out_at` stayed empty).

ID	Requirement	Pri.	Status	Evidence and findings
HMS-F-020	Define fee structures per room type and term	High	**Not implemented**	No fee-structure model; invoice amounts are entered manually.
HMS-F-021	Auto-generate invoice on room allocation	High	**Partially met**	Invoices can be created manually (UT-10). Probe: allocating a bed created 0 invoices.
HMS-F-022	Online fee payment through a payment gateway	High	**Not implemented**	No gateway integration; README states payments are administrative records only.
HMS-F-023	Record offline payments against an invoice and issue a receipt	Medium	**Partially met**	Payments with mode, reference and recorder; balance enforced (UT-11, UT-12, UT-28 to UT-30). No receipt generation; invoice status does not update automatically.
HMS-F-024	Reminder for invoices unpaid near the due date	Medium	**Not implemented**	No reminder job and no email/SMS integration.
HMS-F-030	Student submits complaint with category, description, photo	High	**Partially met**	Complaint with category, description and status Open (UT-13). No photo field; no student-facing form.
HMS-F-031	Warden assigns complaint and updates status	High	**Partially met**	`assigned_to` and status (Open / In progress / Resolved) work (UT-13). Only `updated_at` is stored, not a timestamp per status change; students cannot view status.
HMS-F-032	Notify student on complaint status change	Medium	**Not implemented**	No notification module.
HMS-F-040	Log visitor entry and exit against a student	Medium	**Met**	Visitor log with timestamps and host student; exit-before-entry rejected (UT-16, UT-17, UT-31, UT-32). Entered through the admin console.
HMS-F-041	Leave/outing request with Warden approval and outpass	Medium	**Partially met**	Request with dates, reason, status and reviewer; date validation (UT-14, UT-15). No outpass generation; no student submission form.
HMS-F-042	Mess staff mark daily meal attendance	Low	**Met**	Attendance per student, meal and date; duplicates rejected (UT-18, UT-19).
HMS-F-050	Publish notices to all or selected students	Medium	**Partially met**	Published notices show on the dashboard, drafts do not (UT-03, UT-20). No targeting of a subset; the dashboard is public.
HMS-F-051	Occupancy report per room, floor and block, exportable	Medium	**Partially met**	Dashboard shows active allocation count; admin room list shows occupied and available beds with a block filter. No report and no PDF/CSV export.
HMS-F-052	Fee-dues report, exportable	Medium	**Partially met**	Invoice admin lists paid and balance due with status filter. No export. These columns have no automated test (coverage gap).
HMS-F-053	Create, edit, deactivate users and assign roles	High	**Partially met**	Django users, staff flag and per-model permissions work (probes). The five SRS roles are not pre-defined as groups.

4.2 Non-functional requirements

ID	Requirement	Pri.	Status	Evidence and findings
HMS-NF-001	p95 response time 3 s or less at 200 concurrent users	High	**Not verified**	No load test was run. Not measurable from this environment.
HMS-NF-002	99.5% monthly availability	High	**Not verified**	Requires production monitoring; not applicable to a development build.
HMS-NF-003	SUS usability score of 70 or more	Medium	**Not verified**	No SUS survey with pilot users has been performed.
HMS-NF-004	Scale to 2,000 students and 500 concurrent sessions	Medium	**Not verified**	No capacity test. The default database is SQLite, which the SRS lists as PostgreSQL/MySQL for deployment.

ID	Requirement	Pri.	Status	Evidence and findings
HMS-NF-005	Retain audit logs for at least 3 years	Medium	**Not verified**	Audit events are stored in the database with no purge, but no retention policy is defined or tested.

4.3 Security requirements

ID	Requirement	Pri.	Status	Evidence and findings
HMS-SR-001	TLS 1.2+ mandatory; plain HTTP rejected	High	**Not implemented**	Probe: plain HTTP `/health/` returned 200. `SECURE_SSL_REDIRECT` and secure cookies are not set (see section 5.2). TLS is expected at the Nginx layer in deployment.
HMS-SR-002	Passwords stored with a salted one-way hash	High	**Met**	Probe: stored value uses `pbkdf2_sha256` and contains no plaintext. The SRS names bcrypt/Argon2 as examples; PBKDF2 meets the salted one-way requirement.
HMS-SR-003	RBAC: students cannot reach warden/admin/accounts functions	High	**Partially met**	Per-model permissions return HTTP 403 for staff without rights; a view-only user could not add or open other models. Anonymous and non-staff users are redirected to login (HTTP 302), not 403. No student portal or defined roles.
HMS-SR-004	Session ends after 15 minutes of inactivity	Medium	**Met**	`SESSION_COOKIE_AGE = 900` with `SESSION_SAVE_EVERY_REQUEST = True` (sliding expiry). Configuration verified; the timeout was not exercised end to end.
HMS-SR-005	Tamper-evident audit log of logins, role changes and payments	High	**Partially met**	Admin create/update/delete write audit events with the actor; the audit admin is read-only (add/change/delete all disabled). Login attempts and role changes are not logged; direct database or ORM changes are not logged (signals never registered).

5. Validation Evidence

5.1 Runtime probe results

Probe	Observed result	Finding
Anonymous GET `/` and `/health/`	HTTP 200 for both	**Pass**
Anonymous GET `/admin/` and a model list	HTTP 302 to `/admin/login/`	**Pass**
Non-staff user GET `/admin/` and student list	HTTP 302 to login	**Pass**
Staff user without permissions, student and invoice lists	HTTP 403	**Pass**
Staff with only `view_student`: list / add / room list	200 / 403 / 403	**Pass**
Stored password format	`pbkdf2_sha256`, no plaintext present	**Pass**
Allocate a Pending student to a bed	Rejected: only active students can be allocated a bed	**Pass**
Allocate into a full room	Rejected: this room has no available beds	**Pass**
Set `checked_out_at` on an allocation	Free beds returned to 1	**Pass**
Audit event admin permissions (add / change / delete)	False / False / False (read-only)	**Pass**
Session settings	`SESSION_COOKIE_AGE` 900 s, save on every request	**Pass**
Plain HTTP request to `/health/`	HTTP 200, not redirected or rejected	**Fail**

Probe	Observed result	Finding
Allocation creates an invoice (F-021)	0 invoices created	**Fail**
Mark student Inactive releases bed (F-015)	Bed not released	**Fail**
Direct ORM student create writes audit event	0 audit events (signals not registered)	**Fail**

Probes ran inside Django's test environment against a temporary database. "Fail" means the SRS behaviour was not delivered, not that the probe itself malfunctioned.

5.2 Deployment security configuration

`python manage.py check --deploy` reported 6 warnings. These bear on HMS-SR-001 and on production readiness:

Check	Issue
W004	`SECURE_HSTS_SECONDS` is not set.
W008	`SECURE_SSL_REDIRECT` is not True.
W009	`SECRET_KEY` is the development default (short), unless `HMS_SECRET_KEY` is set.
W012	`SESSION_COOKIE_SECURE` is not True.
W016	`CSRF_COOKIE_SECURE` is not True.
W018	`DEBUG` defaults to True (`HMS_DEBUG` defaults to "1" in `settings.py`).

6. Acceptance Exit Criteria (SRS section 6)

Exit criterion	Assessment	Basis
All High-priority functional requirements implemented and verified	**Not met**	2 of 12 met, 8 partially met, 2 not implemented.
No critical NFR failures remain open	**Not verified**	No NFR test has been run.
All security requirements pass their acceptance tests	**Not met**	SR-002 and SR-004 pass; SR-001 not implemented; SR-003 and SR-005 partial.
RTM shows all associated test cases passed	**Not met**	SRS section 8 lists 8 requirements, all with status N. Suggested updates below.

6.1 Suggested RTM status update

Req ID	Related unit tests	Suggested status	Comment
HMS-F-001	UT-05	**Partially met**	Admin-created records only.
HMS-F-011	UT-08, UT-25 to UT-27	**Met**	Includes the active-student rule (probe).
HMS-F-021	UT-10	**Partially met**	No automatic generation.
HMS-F-030	UT-13	**Partially met**	No photo; no student form.
HMS-F-053	None (probes only)	**Partially met**	Roles not defined as groups.
HMS-NF-001	None	**Not verified**	Load test outstanding.
HMS-SR-002	Probe	**Met**	PBKDF2-SHA256.
HMS-SR-003	Probe	**Partially met**	Redirect instead of 403 for non-staff.

7. Gaps and Recommendations

In order of impact on the SRS exit criteria:

1. **Close the High-priority functional gaps.** Add a fee-structure model and automatic invoice generation on allocation (HMS-F-020, F-021), release beds when a student is marked Inactive (F-015), add a reason field to registration approval (F-003), and either integrate a payment gateway or record it with the guide as out of scope for this release (F-022).
2. **Harden security settings for deployment.** Set `SECURE_SSL_REDIRECT`, `SESSION_COOKIE_SECURE`, `CSRF_COOKIE_SECURE` and HSTS; supply a strong `HMS_SECRET_KEY`; default `DEBUG` to off (HMS-SR-001).
3. **Define roles.** Create Warden, Accountant, Maintenance and Student groups with the right model permissions, and build the student portal so students never use the admin console (HMS-SR-003, F-053).
4. **Complete the audit trail.** Log login attempts (Django `user_logged_in` and `user_login_failed` signals), user and role changes, and fix or remove the unregistered signal module (HMS-SR-005).
5. **Run the non-functional tests.** A load test for HMS-NF-001 and NF-004 (for example with Locust against PostgreSQL), and a SUS survey for NF-003.
6. **Update the RTM** with the statuses in section 6.1 and add the remaining requirements.

8. Limitations

- Validation was performed through Django's test client and source review, not through a browser UI and not on a deployed server.
- Non-functional requirements and the end-to-end session timeout were not measured.
- The Software Test Plan was not available; validation criteria come from the SRS acceptance criteria.
- Manual checks reported earlier by the project owner were not repeated and are not counted as evidence here.

9. Conclusion

The built system meets 6 of the 34 requirements outright, partially meets 15, and does not yet implement 8; 5 non-functional requirements remain unverified. The features that exist are well tested and behave correctly. The main distance to the SRS is missing functionality (student-facing workflows, fee automation, notifications, payment gateway) and deployment security hardening, not defects in what has been built.